



3 (a) Define specific latent heat.

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..... [2]

(b) A dish containing $7.2 \times 10^{-5} \text{ m}^3$ of a substance rests on a laboratory bench. The substance is initially a liquid of density 710 kg m^{-3} . Atmospheric pressure is $1.0 \times 10^5 \text{ Pa}$.

The liquid is heated at its boiling point so that it completely vaporises. The increase in the internal energy of the substance during this process is 17.6 kJ . The final volume of the vapour is 0.017 m^3 .

(i) Show that the magnitude of the work done on the substance when it vaporises is 1.7 kJ .

[2]

(ii) Use the information in (b)(i) to calculate the thermal energy Q , in kJ , supplied to the substance to cause it to vaporise.

$Q = \dots\dots\dots \text{ kJ}$ [2]

(iii) Use your answer in (b)(ii) to determine a value for the specific latent heat of vaporisation L_v , in kJ kg^{-1} , of the substance.

$L_v = \dots\dots\dots \text{ kJ kg}^{-1}$ [2]





(c) The substance in (b) has a specific latent heat of fusion L_F .

Suggest and explain whether L_F is likely to be less than, the same as, or greater than the answer in (b)(iii).

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..... [3]

[Total: 11]

